

DSS 451: Machine Learning Business Applications I
Draft Syllabus ‡

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Class Meetings	MW 5:00 pm-6:00 pm MV 203

Office No.:	Mandeville Hall 349
Office Hours	MW 9am-12pm

Course Description

The course will provide an introduction to Artificial Intelligence (AI) and Machine Learning (ML) applications and methods in Business. A successful professional in the field of Business Intelligence & Analytics (BI&A) is required to be conversant (or at least familiar) with the new era of AI and ML to aid better business decisions.

The course will begin by exploring basic concepts and definitions of the common AI/ML terminology. Once familiarity with these is achieved, we will move on to understanding what AI can and cannot realistically do and what sort of data and analytical methods are commonly used. Some ML methods will also be introduced to the class, followed by additional techniques in the follow-on course DSS 455 (Machine Learning Methods in Business, Spring 2025). Python will be the primary mode of deploying and analyzing Machine Learning Models in class. Frequent use of real-world business case studies may be made in order to help connect these concepts to business applications. Through a capstone project, students will also be able to learn to apply AI/ML to business problems.

This course WILL NOT teach you how to code a programming language. Neither will the focus be on the math behind the algorithms – but instead, we will focus on the applications, analysis and interpretation of machine learning methods.

Learning Objectives

- Understand the basic terminology associated with Machine Learning
- Understand how machines make decisions using algorithms. And which ML algorithm may be appropriate for what situation.
- Using Python to implement ML algorithms
- Improve communication between business owners and data scientists/analysts

Tentative Course Schedule[‡]

Topics	Assignments
Course Introduction and Logistics. What is AI/ML? Why is it important? ML in Business Applications	
Pipeline of Machine Learning Projects Data Representation and Processing, Data Type, Data Separation, Model Evaluation	
Python Mini-bootcamp (Data Preparation)	Assignment 1 (Data Cleaning)
Python Mini-bootcamp (Data Preparation)	
Regression: Linear Regression	Assignment 2 due (Data Cleaning)
Classification: Logistic Regression	Assignment 3 due (Linear Regression)
Classification: Logistic Regression Classification: K Nearest Neighbors (KNN)	Assignment 4 due (Logistic Regression)
Classification: KNN Classification: Decision Tree (CART)	Take-home Mid-term Exam Assignment 5 (KNN)
Classification: Decision Tree (CART) Classification: Random Forest	Assignment 6(Decision Tree)
Classification: Random Forest Classification: Gradient Boosting	Final Project Proposal due
Classification: Gradient Boosting Clustering: K-means	Assignment 7(Random Forest and Gradient Boosting)
Clustering: K-means Variable Selection	Assignment 8 (K-means)
Variable Selection Business Application Case Studies	
Final Presentations	

Electronic Equipment Use

You will use the Python software for the analysis and interpretation of the datasets used in this class. About 50% of each week will be spent working hands-on with Python, applying the algorithms and interpreting the analyses. You will also be assigned homework to be completed in Python workbooks for topics 3-11. Both the Python homework assignments will be timed to occur when a chapter has been completed. Other than these designated times, you are not to use laptops in class. Use of cellphones is not allowed.

Use of Canvas and University Email

Canvas / SJU email will be the preferred method of communication for class-related announcements, events, changes, etc. It is the student's responsibility to check Canvas and his/her University email account regularly. Instructor is not responsible for using email accounts other than the student's assigned University account.

Course Policies

Pre-Requisites

- DSS 420
- DSS315 (Python) is required either as a pre-requisite or a 'co-requisite'

Required Course Materials

- Instructor will provide slides, datasets and Jupyter notebooks required for the course
- Software: Python (Free and Open Source; Installation instructions provided in class)

Optional but recommended texts (in order of importance):

- Introduction to Machine Learning with Python: A Guide for Data Scientists, Müller & Guido
 - Free e book from library <https://catalog.sju.edu/record=b1430809~S9>
 - There is 1 copy of the ebook. It can be read online or checked out (1 day use - it would be similar to having a book copy on library reserve for overnight check out). <https://guides.sju.edu/ebookdownloads>
- Machine Learning For Dummies, Mueller & Massaron
- Machine Learning For Dummies®, IBM Limited Edition
<https://www.ibm.com/downloads/cas/GB8ZMQZ3>
- The Master Algorithm: How the Quest for the Ultimate Learning Machine Will Remake Our World by Pedro Domingos - Read for pleasure
- 60+ free books: <https://www.kdnuggets.com/2015/09/free-data-science-books.html>

If a bit of Math doesn't scare you

- The hundred-page machine learning book (<http://themlbook.com/>) – Pay what you want
- An Introduction to Statistical Learning (with applications in R) Free (www.statlearning.com)

Other readings as assigned by instructor and/or posted on the course schedule and Canvas site.

Course Grading

Final course grade will be assigned based on the following scale. Please note that *extra-credit work is not available at any point for this course since there are no weekly assignments.*

Assignment	35%
Mid Term Exam	25% (Approx. Week 8)
Final Project	35% (Instructor evaluation (70%) and peer evaluation (30%))
Class Participation	5% (Attendance, adhere to class policies)
TOTAL	100 points

Final Grade Distribution

A	95-100 points	C+	77-79 points
A-	90-94 points	C	73-76 points
B+	87-89 points	C-	70-72 points
B	83-86 points	D	60-69 points
B-	80-82 points	F	0-60 points

Note: Points round up to the nearest 0.5. (Eg: 86.4 is a B, 86.6 is a B+)

Make-up Policy

Students are expected to be present for all in-class quizzes, examinations as listed on the syllabus or announced by the instructor. Make-up exams require valid documentation (this includes medical and other family emergencies). Valid documentation of an absence must be requested through the Dean's Office and presented to the instructor in order to reschedule an exam. ***There will be no exceptions made to this policy.*** Please do not request makeup exams or extended assignment deadlines based on family / leisure / travel.

Consult the University's Academic Calendar and plan accordingly. *Missed quizzes / exams will result in a grade of zero (0) points for that exam.*

Responsible Employee Reporting

While I want you to feel comfortable coming to me with issues you may be struggling with or concerns you may be having, please be aware that I have some reporting requirements that are part of my job responsibilities at Saint Joseph's University. For example, if you inform me of an issue of sexual harassment, discrimination or sexual misconduct (Sexual Assault, Sexual Harassment, Sexual Exploitation, Domestic Violence, Dating Violence, or Stalking) I will keep the information as private as I can, but I am required to bring it to the attention of the institution's Title IX Coordinator, Mary-Elaine Perry. If you would like to speak with her directly, she can be reached at mperry01@sju.edu or 610-660- 1145.

Counseling and Psychological Services (CAPS) can provide confidential support and can assist you with determining how you would like to address this issue. CAPS is located at 504A Merion Gardens and can be reached at 610-660-1090. They also have walk-in hours M-F 11- 12:30 in LaFarge. REPP, Rape Education and Prevention Program provides a 24-hour confidential helpline (610-733-9650) as well. For more information about your confidential options as well as community resources go to Support Website.

Attendance

Attendance and participation are essential for learning. Therefore, attendance is required for all in- person classes and synchronous online sessions. Unexcused absences are likely to result in course penalty; more than 4 unexcused absences may result in a grade of FA (Failure due to Absence) for the course.

If you are unable to attend class due to a non-COVID-19 related illness, communicate directly with me through email. If you have COVID-19 symptoms, test positive or are exposed to an unknown or non-

SJU individual, you must complete the COVID-19 Self-Disclosure Form. If the University places you in quarantine or isolation I will be notified. If you don't notify me through email and I do not receive a COVID-19 notification from the University, you will be marked absent.

If your physical absence is due to isolation or quarantine, it is your responsibility to contact me within 48 hours outlining both the date of your expected return to class and a plan to stay on pace with readings and assignments during your mandated physical absence.

Email Response Time:

During the week, emails will be returned within 24 hours. Any emails sent on weekends will be returned by Sunday evening if not before.

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Students With Disabilities:

Reasonable academic accommodations may be provided to students who submit appropriate documentation of their disability. If students have need of assistance or questions with this issue, they are encouraged to contact the Office of Student Disability Services (SDS) at sds@sju.edu or by phone at 610.660.1774. The Office of SDS also provides an appeal/grievance procedure for complaints regarding requested or offered reasonable accommodations. More information can be found at: www.sju.edu/sds.

Class Recording:

Be advised that by taking this course you consent to some or all class sessions being recorded by the instructor (and in some cases with assistance by the SJU Office of Information Technology). Recordings are intended to focus on the instructor but the voices and/or likenesses of other individuals in the class may also be recorded. Video and/or sound recordings will be stored at a secure SJU video management platform for two years and will be used or distributed only for educational or academic purposes, e.g., to allow for remote learning for those students not physically present in class or unable to attend a synchronous online class session. If you have questions about the use of such technology or you do not consent to being recorded, you must discuss and resolve your concerns with your instructor before the end of the drop/add period.

Bias, Discrimination, & Harassment:

As a Catholic, Jesuit University, Saint Joseph's is committed to providing a workplace and educational environment, as well as other benefits, programs, and activities, that are free from acts of bias, discrimination, harassment, including acts of sexual harassment, sexual assault, dating and domestic violence, and stalking. We encourage anyone who has experienced this type of harm to seek help from the University by filing a report through the incident reporting form, or by calling 610-660-1145.

For information about the University's response to reports of bias, harassment, and discrimination, including definitions of prohibited conduct, information regarding confidential reporting options on- and off-campus, and resources for support, please visit <https://www.sju.edu/offices/titleix-equity> or contact titleix@sju.edu / bias@sju.edu.

While I want you to feel comfortable coming to me for support, please know that I have some reporting requirements that are part of my job responsibilities at Saint Joseph's University. For example, I am required to report any student disclosures of bias, discrimination, and harassment. I will keep the information as private as I can, but I am required to bring it to the attention of the institution's Director Office of Title IX, Equity & Compliance.

Anonymous Student Feedback

The following link allows students to provide anonymous feedback directly to the Decision & System Sciences Department (DSS). Students who believe the situation they are experiencing warrants immediate attention can notify the Department via this anonymous survey. Students may also email the Department Chair (Dr. Virginia Miori) directly at vmiori@sju.edu. Please note that this link is not meant to be used as a

faculty evaluation and does not replace the student evaluation at the end of the semester.

https://sju.co1.qualtrics.com/jfe/form/SV_4NSi1TCKdsHWF5X

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Academic Honesty:

Saint Joseph's University encourages the free and open pursuit of knowledge; we consider this to be a fundamental principle and strength of a democratic people. To this end, SJU expects its students, its faculty, its administrators, and its staff to uphold the highest standards of academic integrity. The University expects all members of the University community to both honor and protect one another's individual and collective rights.

The SJU Academic Honesty Policy, which is published both in the Student Handbook and in the University Catalog, will be strictly enforced. It is your responsibility to understand this policy. Use of unauthorized notes or assistance during an exam will result in at least failure of the assignment, if not failure of the course. Plagiarism will also result in at least failure of the assignment, if not failure of the course. Plagiarism occurs in any instance where one submits as one's own work statements or ideas taken from another source but not properly acknowledged as such. Any

material copied or derived from another source must be acknowledged by giving a reference. For direct quotes, always use quotation marks and provide a reference. If you paraphrase something, you must provide a reference to acknowledge where you found the material. Internet sources, if used, must be cited.

Students are expected to include the following Academic Honesty Affirmation on each of their assignments: "I attest to the honesty of my academic work and affirm that it conforms to the standards of the Academic Honesty Policy of Saint Joseph's University."

AI Usage Policy for Academic Integrity:

Purpose: This policy is designed to clarify the expectations and limitations regarding the use of Artificial Intelligence (AI) tools in completing coursework for this class.

Scope:

- AI tools include, but are not limited to, machine learning models, chatbots (e.g., ChatGPT), AI content generators, and automated problem solvers.
- This policy applies to all assignments, quizzes, exams, and projects unless otherwise specified.

Guidelines:

- **Disclosure:** Students must disclose the use of any AI tools in completing coursework. Each submission should include a statement or footnote indicating the tools used and the extent of their involvement.
- **Prohibited Uses for Exams and Quizzes:** AI tools are not permitted for responding to open-ended questions or expressing personal opinions and thoughts during exams and quizzes. However, you may use these tools for assistance with coding tasks.
- **Original Work:** AI should not be used to generate entire essays, reports, or any other assignment meant to assess individual thought and analysis. Students are encouraged to use their own reasoning and capabilities.

Permitted Uses: Research Assistance: AI can be used for gathering background information, data analysis, and summarization to aid understanding.

- **Drafting Assistance:** AI may be used to help draft outlines or to provide suggestions for improving grammar and style in early drafts. However, the final submission must significantly reflect the student's original work and understanding.
- **Coding Assistance:** AI may be used to help debug the code and draft the template of the code.

Ethical Considerations: Students must ensure that the use of AI does not infringe on the rights of others or violate any laws regarding data privacy and intellectual property.

Submissions generated with the help of AI must adhere to the principles of academic honesty and integrity.

Consequences: Failure to comply with this policy will be considered a violation of academic integrity and will be subject to disciplinary actions as outlined in the student handbook.

Discussion and Updates:

- Students are encouraged to discuss the use of AI tools with the instructor if unsure about appropriate applications.
- This policy may be updated as new AI technologies and applications emerge.

Changes to Syllabus:

This syllabus is subject to change at the sole discretion of the instructor. Any changes will be announced class- wide in a timely manner and posted on the class website.

Inclusivity Statement:

I am committed to providing an atmosphere for learning that respects diversity. While working together to build this community I ask that we all:

- Recognize that our classroom community represents a rich variety of backgrounds and perspectives.
- Share our unique experiences, values, and beliefs.
- Be open to the views of others.
- Honor the uniqueness of our community.
- Appreciate the opportunity that we have to learn from each other in this community.
- Value each other's opinions and communicate in a respectful manner.
- Keep confidential discussions that the community has of a personal (or professional) nature.
- Use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the SJU community.

Food & Basic Needs

Your well-being is of primary importance to the learning environment. Facing financial challenges, such as difficulty purchasing textbooks, technology or other obstacles -- like securing food, basic needs, or transportation -- may impact your ability to succeed in your courses.

If you are facing any of these hardships, please complete this form, which will be sent to the Office of Student Success. A staff member will work with you to connect you with University and/or other resources. For support with food and basic needs, you are encouraged to visit HawkHUB in person or submit an online order.

Finally, you are welcome to contact me to discuss specific ways in which I can support you. I am

committed to helping you succeed.